

John Doe

MSc Aerospace Engineering - Research Assistant - AIAA Student Member

☎ +41 79 123 45 67 ✉ john.doe@unihelios.ch 🌐 johndoe.dev 🔗 linkedin.com/in/johndoe 🐙 github.com/johndoe
📍 Bern, CH

“ Aerospace Engineering graduate student at University of Helios, passionate about computational fluid dynamics and sustainable aviation. ”

EDUCATION

- University of Helios — MSc Aerospace Engineering** 2023 – present
Focus: Computational Fluid Dynamics · GPA 5.8 / 6.0
- Polytechnic Institute of Varda — BSc Mechanical Engineering** 2019 – 2023
Focus: Thermodynamics and Fluid Mechanics · GPA 1.2 (German grading system)
Thesis: *Numerical Simulation of Turbulent Flow over a NACA 0012 Airfoil*
- Gymnasium Solaris — Abitur** 2019
Final grade: 1.0 · Valedictorian · Awarded distinction in Mathematics and Physics

EXPERIENCE

- University of Helios — Research Assistant** 2024 – present
Institute of Fluid Dynamics
Supporting research on high-speed compressible flows in rotating detonation engines. Developing CFD simulation pipelines in Python and OpenFOAM. Assisting with experimental test campaigns and data analysis.
- Aerovaunt AG — Engineering Intern** Summer 2022
Aerodynamics Department, Bern
Contributed to aerodynamic shape optimisation workflows using simulation tools and MATLAB. Automated post-processing of simulation results, reducing analysis time by 40%.
- Polytechnic Institute of Varda — Teaching Assistant** 2021 – 2023
Thermodynamics I and II
Guided undergraduate students through course material and weekly exercise sessions for two consecutive semesters.

PROJECTS

- Autonomous UAV for Search and Rescue** 2023
Student project — Python, ROS2, Fusion 360
Designed and built a fixed-wing UAV capable of autonomous waypoint navigation. Implemented a custom flight controller in Python using sensor fusion and PID control. Achieved 45-minute endurance in field tests. [🔗 github.com/johndoe/sar-uav](https://github.com/johndoe/sar-uav)
- CFD Surrogate Model for Airfoil Optimisation** 2024
Research project — PyTorch, OpenFOAM
Trained a neural network surrogate model to predict lift and drag coefficients from airfoil geometry parameters, achieving 98% accuracy at 1000× speedup over full CFD simulations.

AWARDS & SCHOLARSHIPS

- Helix Foundation Scholarship** 2023 – present
Awarded to the top 1% of engineering students nationwide
- Varda Student Paper Competition — 1st place** 2023
Polytechnic Institute of Varda Annual Research Awards
- Solaris Physics Prize** 2019
Awarded by Gymnasium Solaris for outstanding performance in Physics
-

PUBLICATIONS

Surrogate Modeling for Transonic Airfoil Optimisation Using Deep Neural Networks

J. Doe, M. Mustermann, A. Smith

Journal of Computational Aerodynamics - peer-reviewed - [doi: 10.0000/example](#)

2024

Numerical Investigation of Turbulent Boundary Layer Separation on Swept Wings

J. Doe, M. Mustermann

International Journal of Fluid Engineering - under review

2023

SKILLS

Python

C++

OpenFOAM

Ansys Fluent

PyTorch

MATLAB

ROS2

Fusion 360

Siemens NX

LaTeX

Typst

Git

LANGUAGES

English - native

German - C2

Spanish - B2

French - A2

Bern, June 2026